CLASS 10 · CBSE · NCERT MATHEMATICS · FORMULA BANK

Chapter 9: Some Applications of Trigonometry

Every formula and result of the chapter on one sheet

1	Angle of elevation	Angle between the horizontal and the line of sight, object above eye level
2	Angle of depression	Angle between the horizontal and the line of sight, object below eye level
3	Elevation and depression	Angle of depression of B from A = angle of elevation of A from B
4	Height and ground distance	$\tan\theta = \frac{\text{height}}{\text{distance}}$, so $\text{height} = \text{distance} \times \tan\theta$
5	Height and slant length	$\sin\theta = \frac{\text{height}}{\text{slant length}}$
6	Ground distance and slant length	$\cos\theta = \frac{\text{distance}}{\text{slant length}}$
7	tan of standard angles	$\tan 30^\circ = \frac{1}{\sqrt{3}}, \tan 45^\circ = 1, \tan 60^\circ = \sqrt{3}$
8	sin of standard angles	$\sin 30^\circ = \frac{1}{2}, \sin 45^\circ = \frac{1}{\sqrt{2}}, \sin 60^\circ = \frac{\sqrt{3}}{2}$
9	cos of standard angles	$\cos 30^\circ = \frac{\sqrt{3}}{2}, \cos 45^\circ = \frac{1}{\sqrt{2}}, \cos 60^\circ = \frac{1}{2}$
10	Complementary angles	$\tan(90^\circ - \theta) = \cot\theta$, so $\tan\theta \times \tan(90^\circ - \theta) = 1$
11	Useful values	$\sqrt{2} \approx 1.414, \sqrt{3} \approx 1.732$